

MITOCHONDRIAL PREPROTEIN IMPORT: TIM50 MEDIATED COUPLING OF PRESEQUENCE TRANSLOCASE AND RESPIRATORY CHAIN COMPLEXES

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ABSTRACT

Biogenesis of mitochondria, organelles crucial for cell metabolisms and signaling, strictly depends on the import of >1000 protein precursors (preproteins) (Moulin et al., 2019). The mitochondrial membrane potential ($\Delta\Psi$) generated by respiratory chain complexes (RCCs) energizes consecutive preprotein import steps mediated by the presequence translocase (TIM23). In addition to promote electrophoretic transport of cleavable, positively charged presequence signals, $\Delta\Psi$ also energizes transport of mature protein segments. The translocase accessory subunits Tim21 and Mgr2 mediate an association of TIM23 with RCCs thus maintaining optimal $\Delta\Psi$ during protein sorting into the membrane (Gebert et al., 2012). In addition, Pam17 promotes import of matrix targeted preproteins that are particularly dependent on $\Delta\Psi$ (Schendzielorz, 2017), however it is unclear if matrix-translocating TIM23 physically associates with RCC.

We have recently shown that the TIM23 core subunit Tim50 may also interact with RCCs (Caumont-Sarcos et al., 2020), but the molecular basis and functional significance of this interaction have remained unclear. Here, using affinity purification, blue native PAGE and mitochondrial import assays in wild-type and mutant yeast mitochondria, we investigated the molecular basis and functional relevance of the Tim50-RCC interaction. We found that Tim50 binds to mature respiratory chain supercomplex III_2IV and III_2IV_2 . This interaction is independent of the known TIM23-RCC coupling mediated by Tim21 and Mgr2, and is enhanced upon deletion of the Tim50 IMS domain. Of note, our data indicate that this new Tim50-RCC coupling mechanism promotes import of matrix-targeted preproteins particularly dependent on $\Delta\Psi$. We conclude that distinct TIM23-RCC coupling mechanisms contribute to mitochondrial preprotein import.

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